INCH-POUND

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29 June 90

COMMERCIAL ITEM DESCRIPTION

TEST SET, INTERFACE BUS

The General Services Administration has authorized the use of this Commercial Item Description (CID).

Abstract. This CID describes the minimum requirements for a Test Set, Interface Bus used to test parallel/BCD, IEEE-488, RS-232-C, and CIIL electronic test interfaces. Equipment covered by this CID will be commercially available equipment modified to the extent necessary to meet the following description.

Salient Characteristics. The equipment shall be capable of operation and be bid sample tested within the accuracies, limits, and specifications herein.

1. Safety and Environmental. The equipment shall be UL (Underwriters Laboratories) approved and meet established safety requirements.

2. Electrical Power Sources. The equipment shall operate from nominal commercial, military, and shipboard power sources of 100, 110, 120, 200, 220 and 240 volts AC (+ or - 10%) at line frequencies of 50 to 60 Hertz.

3. Temperature Requirements. In an operating environment, this equipment shall meet performance and accuracy requirements between 0 and +40 degrees Celsius. When stored or nonoperating, the equipment shall withstand temperaature of -10 to +50 degrees Celsius.

4. Mean Time Between Failure (MTBF). The equipment shall function with a Mean Time Between Failure Rate of 2,000 hours. The lower MTBF shall be 2,000 hours and the upper test MTBF shall be 4,000 hours.

Beneficial comments, recommendations, additions, deletions, etc. or any data which may improve this document should be sent to: Technology and Industrial Services Division, SA-ALC/TIRDM, Kelly AFB TX 78241-5000.

AMSC N/A FSC 6625

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5. Pre-conditioning Requirement. Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn-in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

6. Calibration and Maintenance Adjustments. The design of the test set shall provide for readily accessible calibration adjustments and maintenance adjustments. These adjustments shall be provided by variable value components which are adjustable by use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater, based on an operating time of 2,000 hours.

7. Performance Characteristics. The Test Set, Interface Bus shall include, but not be limited to, the following equipment and application testing:

a. Computing instrument Controller shall include:

(1) Program development and system software package shall utilize ATLAS 716C;

(2) Basic interpreter;

(3) 0.7 MIPS processor;

(4) Macrostore high-speed floating point processor;

(5) Detachable programmer's keyboard;

(6) Minimum of 2 expansion slots;

(7) 400 Kilobyte floppy disk drive;

(8) 40 Megabyte internal hard drive;

(9) Real time non-volatile calendar/clock;

(10) One Megabyte RAM expansion;

(11) Interfaces: parallel/BCD, IEEE-488, RS-232-C and the ability to test CIIL instruments;

(12) Printer Interface: RS-232-C

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b. CRT Display. The display shall be a high resolution, high contrast graphic display which is touch sensitive operationally or operates interactively via keyboard function keys. The display format shall be selectable for 16 lines of 80 characters or 8 lines of 40 double-size characters.

c. Scanning method. The CRT shall utilize a non-interlaced raster scan.

d. Character memory. There shall be at least 1280 bytes of dedicated display memory.

e. Character set shall contain a minimum of 96 standard ASCII characters including math characters and graphics characters.

f. Enhancements. The CRT display shall contain reverse video and high-lighting character enhancements.

g. Cursor. The cursor shall be selectable for block or suppressed.

h. Keyboard. The programmer's keyboard shall be QWERTY style, detachable unit, with full ASCII characters set plus required function keys. Keyboard shall be designed to withstand damage when placed upside down.

i. IEEE-488 Interface Handshake Repertoire shall contain the following functions:

(1)	Source Handshake	SH1
(2)	Acceptor Handshake	AH1
(3)	Basic Talker	T6
(4)	Basic Listener	L4
(5)	Controller	C1 thru C4, C6
(6)	Device Trigger	DT1
(7)	Device Clear	DC1
(8)	Service Request	SR1
(9)	Extended Talker	TE
(10)	Extended Listener	LE
(11)	Remote/Local	RL

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j. IEEE-488 Remote Message Capability shall communicate with instruments using the following functions:

(1)	Attention	ATN
(2)	Data Byte	DAB
(3)	Data Accepted	DAC
(4)	Data Valid	DAV
(5)	Device Clear	DCL
(6)	Interface Clear	IFC
(7)	My Listen Address	MLA
(8)	My Talk Address	MTA
(9)	Remote Enable	REN
(10)	Ready for Data	RFD
(11)	Unlisten	UNL
(12)	Untalk	UNT

k. RS-232-C Interface. The controller shall include an RS-232-C interface with the following resettable parameters for each RS-232-C port :

(1) Baud Rate (75 - 19,200 baud)

(2) Data Bits per character

(3) End of file character

(4) End of line character

(5) Parity

(6) Stop bits

(7) Stall input enable/disable (XON/XOFF)

(8) Stall output enable/disable (XON/XOFF)

(9) Time out

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l. Parallel Interface. The controller shall be able to bidirectionally transfer monitor and control status bits, 16-bit words, or 512-word blocks via parallel interface. The following handshake or strobe protocols may also be selected:

(1) Status Input/Output

(2) Full handshake input and output

(3) Pulse handshake input and output

(4) Strobe input and output

m. Real-time clock. The controller shall be supplied with a battery back-up, real-time clock. The clock shall supply year, month, day, hour, minute, and second information with leap-year handling. The clock accuracy shall be a minimum of 3 sec/day.

n. Support Equipment. The controller shall contain all support equipment required for testing purposes.

o. Documentation. The controller shall include all development and testing documentation for support.

p. Hard Disk Drive. The bus interface test set shall include a minimum 40 Megabyte hard disk drive. The disk drive shall have a transfer rate of 160 Kilobyte per second with on-line capacity of 40 Megabytes.

q. Printer. The test set's printer shall be a dot matrix or ink jet performing at a speed of 100 characters/second or better and capable of operating bi-directionally. The paper size shall be 4.5 to 10.0 inches. The interface shall be Serial RS-232 with band rates from 110 to 4,800 bits/second. The minimum buffer capacity shall be 2,000 characters with XON/XOFF protocol.

r. Manuals. The test set shall include two sets of operating, programming, instruction, and documentation manuals.

s. Cables. The test shall include one RS-232 interface cable (2M long), one serial printer cable (2M long), and two IEEE-488 interface cables (2M long).

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3. Contractor Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the characteristics of this CID, and that the product conforms to the manufacturer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery, and thereafter, as may be otherwise provided for under the provisions of the contract.

4. Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using Inch-Pound units, provided they fall within specificed tolerances using conversion tables contained in the latest revision of Federal Standard No. 376 and all other requirements of this CID are met.

5. Regulatory Requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

6. Preservation Packaging, Packing, Labeling, and Marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Notes.

FED-STD 376 Preferred Metric Units for General Use by Federal Government

This document is available from the Defense Printing Service and Detachment Office, Bldg. 4D, NPM-DODSSP, 700 Robbins Avenue, Philadelphia, PA 19111.

The following product is known to have previously met requirements of this CID and does not require resubmission of bid samples.

Manufacturer	Model	National Stock Number
Fluke	A1002	6625-01-251-1206

Custodian:
Air Force - 99
Navy - SH
DLA - ES

Preparing Activity:
Air Force - 82

Project 6625-0733

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